

PROJECT SYNOPSIS

1. Project Title

Design and Development of a Low-Power, High-Speed FPGA-Based System for Real-Time Epileptic Seizure Detection

2. Objective

The primary objective of this project is to design and implement a lightweight, hardware-accelerated epileptic seizure detection system on a Field-Programmable Gate Array (FPGA). The system aims to process digitized Electroencephalography (EEG) signals in real-time to detect the onset of seizures with high accuracy, minimal latency, and extremely low power consumption, making it suitable for wearable biomedical applications.

3. Problem Statement

Epilepsy affects approximately 50 million people worldwide. Real-time detection of seizures is critical for patient safety and timely medical intervention. Conventional software-based detection systems (using CPUs/GPUs) suffer from high power consumption, processing latency, and bulky hardware, making them unsuitable for continuous, battery-operated wearable monitoring. There is a need for a dedicated hardware solution that can perform continuous monitoring with high deterministic speed and low power.

4. Proposed Methodology

The project employs a custom digital logic design using Verilog HDL, implementing a **Sliding Window Spike Counter** algorithm.

- **Data Acquisition:** Real patient EEG data is sourced from the CHB-MIT Scalp EEG Database and quantized into 16-bit fixed-point test vectors.

- **Algorithm Execution:** The FPGA continuously processes the incoming samples at a biological rate of 256 Hz. It applies an amplitude threshold ($\pm 200 \mu\text{V}$) to identify abnormal neural spikes.

- **Temporal Windowing:** A 512-sample (2-second) sliding window continuously tracks the spike count. If the accumulated spikes exceed the threshold of 25 within the window, a seizure is officially flagged.

- **Hardware Mapping:** The algorithm is synthesized and deployed onto the Xilinx Artix-7 FPGA, utilizing Block RAM for data storage and custom clock dividers for accurate sample timing.

5. Hardware & Software Requirements

Hardware Requirements:

- Digilent Basys 3 FPGA Trainer Board (Xilinx Artix-7 xc7a35tcpg236-1)
- Micro-USB Cable (for programming and power)
- Modern PC for synthesis and simulation

Software Requirements:

- Xilinx Vivado ML Edition (for Synthesis, Implementation, and Bitstream generation)
- Python 3.x (with `mne` and `numpy` libraries for initial EEG data extraction)
- Verilog HDL (Hardware Description Language)

6. Expected Outcomes and Results

The system has been successfully verified through both behavioral simulation (XSim) and physical hardware deployment. Testing against a 5-minute EEG recording containing both normal activity and a 40-second seizure event demonstrated:

- The system will demonstrate successful detection of seizure events during testing with minimal false positives.
- **Reliability:** Zero false positives during normal, inter-ictal brain activity.
- **Efficiency:** The design consumes less than 3% of the available Artix-7 logic resources, proving its extreme lightweight nature and suitability for edge-computing medical devices.

Future enhancements may include machine learning-based classification and integration with IoT-enabled healthcare monitoring systems

7. Scope of the Project

This project focuses on implementing a real-time epileptic seizure detection algorithm using FPGA hardware acceleration for wearable healthcare applications. The work includes EEG signal

preprocessing, spike detection, sliding window analysis, and FPGA deployment using Verilog HDL. The system is intended for research and educational purposes and does not replace certified clinical diagnostic systems.

References

- [1] M. Valentino et al., "Wearable Epilepsy Seizure Detection on FPGA with Spiking Neural Networks," **IEEE Trans. on Biomedical Circuits and Systems**, DOI: 10.1109/TBCAS.2025.3575327, 2025.
- [2] World Health Organization, "Epilepsy," WHO Fact Sheet, Feb. 2024.
- [3] AMD/Xilinx Inc., "Artix-7 FPGAs Data Sheet: DC and AC Switching Characteristics," DS181, v1.27.1, Jul. 2024.
- [4] S. Kumar et al., "FPGA implementation of automatic seizure detection in EEG signals using machine learning algorithm," **Discover Applied Sciences**, vol. 6, no. 383, 2024
- [5] M. Sahani, S. K. Rout, and P. K. Dash, "FPGA Implementation of Epileptic Seizure Detection Using Semisupervised Reduced Deep Convolutional Neural Network," **Applied Soft Computing**, vol. 110, 107639, 2021.
- [6] M. H. KN, "Hardware acceleration of high sensitivity power-aware epileptic seizure detection system using dynamic partial reconfiguration," **IEEE Access**, vol. 9, pp. 75071-75081, 2021
- [7] C. Peng et al., "FPGA-based acceleration for Bayesian convolutional neural networks for seizure detection," **IEEE Trans. on Biomedical Circuits and Systems**, vol. 14, no. 4, pp. 624-634, 2020.
- [8] I. Ullah et al., "An automated system for epilepsy detection using EEG brain signals based on deep learning approach," **Expert Systems with Applications**, vol. 107, pp. 61-71, 2018.